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October 6, 2026

Results Report

Weather Project

The purpose of this project is to use the data science methodology to determine whether it rains more in Seattle, WA or Pittsburgh, PA.

For this project, I used weather data for Seattle and Pittsburgh from the National Oceanic and Atmospheric Administration (NOAA). The data used in the analysis covers January 1, 2018 through December 31, 2022. During the data cleaning step, the date range used for the analysis was changed to November 11, 2018 through December 31, 2022.

The data from this source includes precipitation, date, and location information needed for the analysis.

For this project, I consider a city to be rainier if it has precipitation on a greater number and proportion of days. Both cities are known for rainy weather, so I was genuinely curious to see which city has more rainy days based on the data.

Analysis

First, I needed to inspect the data from Seattle and Pittsburgh. I checked the columns in both datasets and made sure that I had date and precipitation columns. Then I compared the dataset sizes and checked if I had enough data for my analysis.

I converted the date column to datetime and checked for duplicates. Both datasets did not have duplicate dates. Then I saw that the Pittsburgh dataset did not have data before 11-04-2018, and the Seattle dataset was missing data from 11-04-2018 to 11-11-2018. This is why I decided to limit the date range from 11-11-2018 to 12-31-2022.

The next step was to join both datasets. In the new dataset, I had date, city, and precipitation columns. I started to inspect the newly created dataset. There were 205 missing precipitation values: 118 for Seattle and 87 for Pittsburgh. I used mean precipitation values to fill in the missing data.

After the dataset was clean, I saved it and started the analysis.

I decided to analyze mean daily precipitation, the number of days with precipitation, and the proportion of days with precipitation.

Mean daily precipitation is significantly higher in Seattle in January, November, and December.

Mean daily precipitation

is significantly higher

in Pittsburgh in April,

May, July, and August.

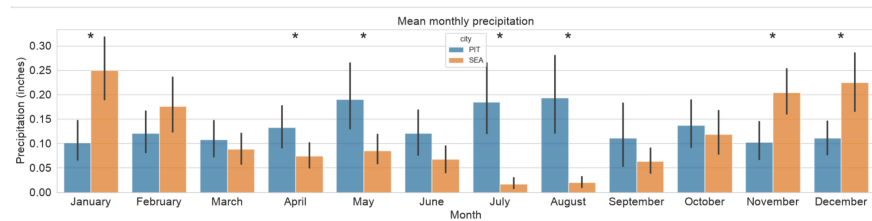
No statistically

significant difference

was detected in

February, March, June,

September, or October. The results are similar.



The proportion of days

with precipitation is

significantly higher in

Seattle in January,

February, November,

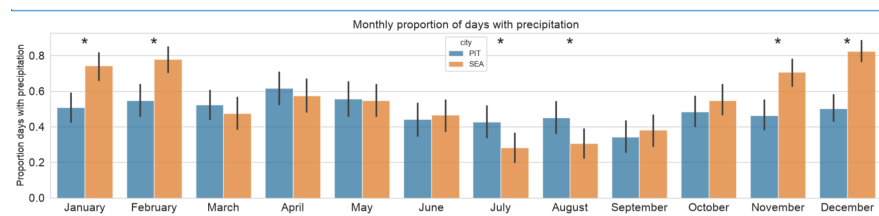
and December. It is

significantly higher in

Pittsburgh only in July

and August. No statistically significant difference was detected in March, April, May, June,

September, or October.



Here, the results are more visible. This is why I decided to use the proportion and number of days with precipitation as the main criteria for my conclusion.

Conclusion

Based on my analysis, both Seattle, WA and Pittsburgh, PA are rainy cities, and I noticed that they are actually very similar.

The mean daily precipitation values are very close. Seattle has a mean daily precipitation of 0.1189 inches, while Pittsburgh has 0.1339 inches.

I decided to determine which city is rainier based on the number and proportion of days with precipitation. Seattle had more days with precipitation, with 846 days compared with 739 days in Pittsburgh.

The proportion of days with precipitation is significantly higher in Seattle in January, February, November, and December. It is significantly higher in Pittsburgh in July and August.

Based on the number and proportion of days with precipitation, I conclude that Seattle is rainier than Pittsburgh.